

Machine Learning Project Neue Fische

06/05/2026

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DATA: Overview

Kaggle link: <https://www.kaggle.com/competitions/playground-series-s6e4/overview>

Goal: **Predicting Irrigation Need** (Low, Medium, High)

Dataset: Precleaned, synthetically generated from real-world data

Shape (training data): 20 features x 630.000 observations

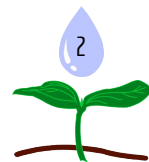
Features:

Categorical Features

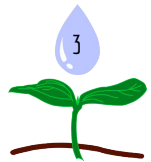
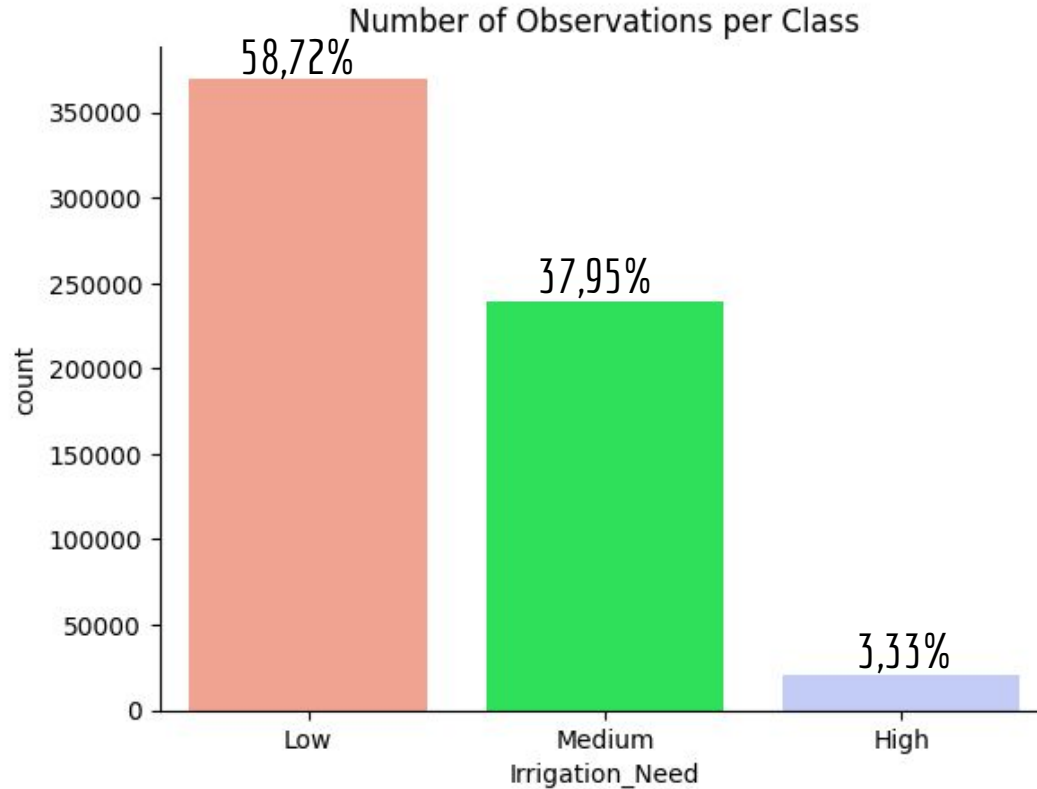
- Soil Type
- Crop Type
- Crop Growth Stage
- Season
- Irrigation Type
- Water Source
- Region
- Mulching Used

Numerical Features

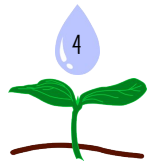
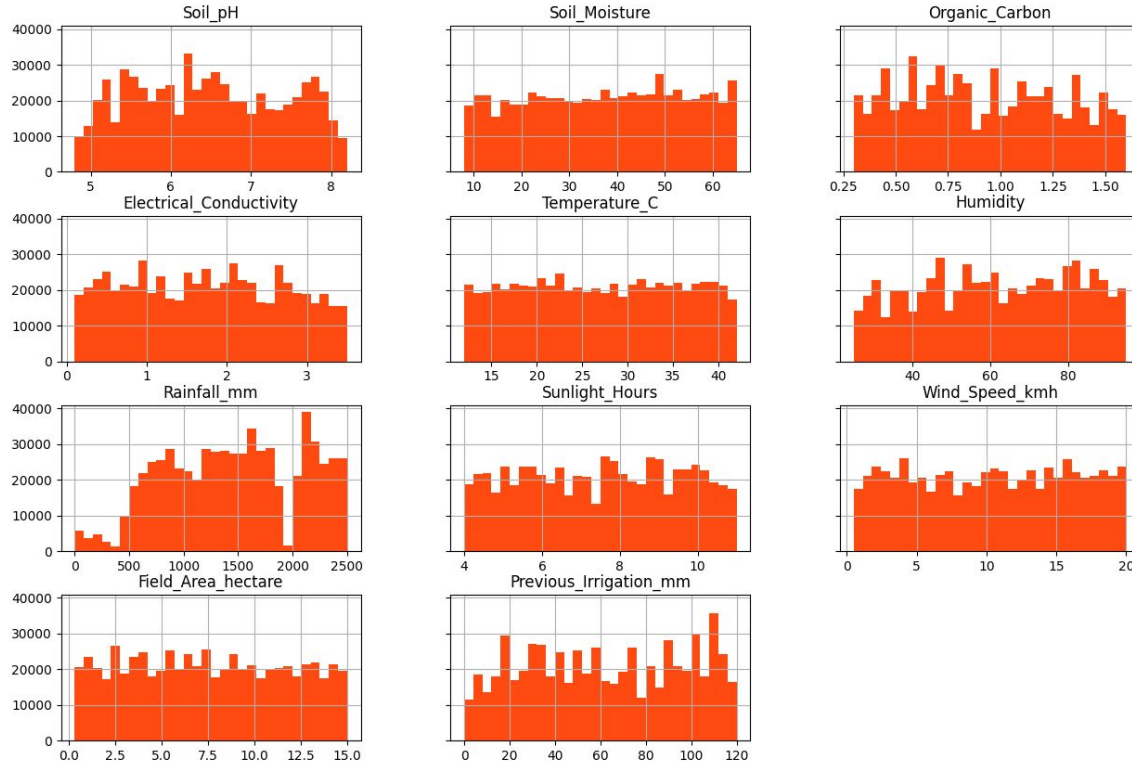
- Soil pH
- Soil Moisture
- Organic Carbon
- Electrical Conductivity
- Temperature
- Humidity
- Rainfall mm
- Sunlight Hours
- Wind Speed
- Field Area hectare
- Previous Irrigation



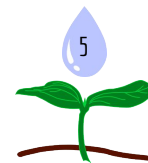
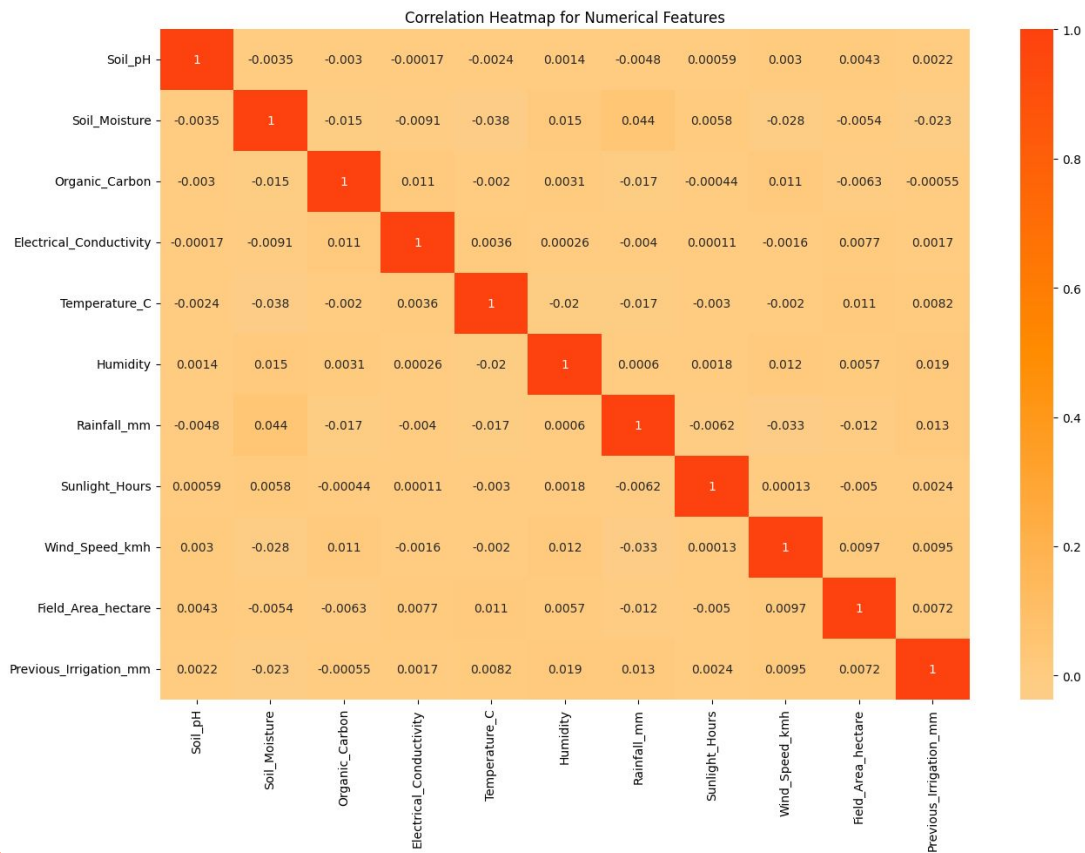
EDA: Target Variable



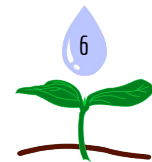
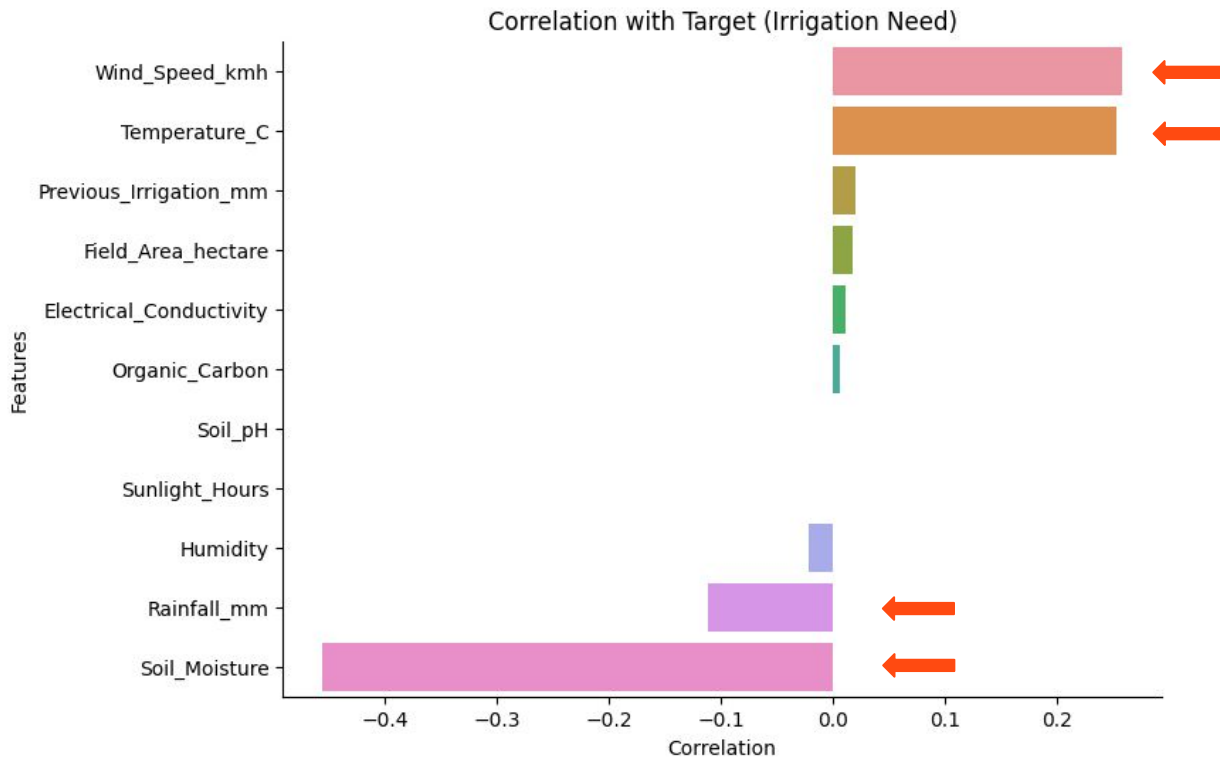
EDA: Numerical Features



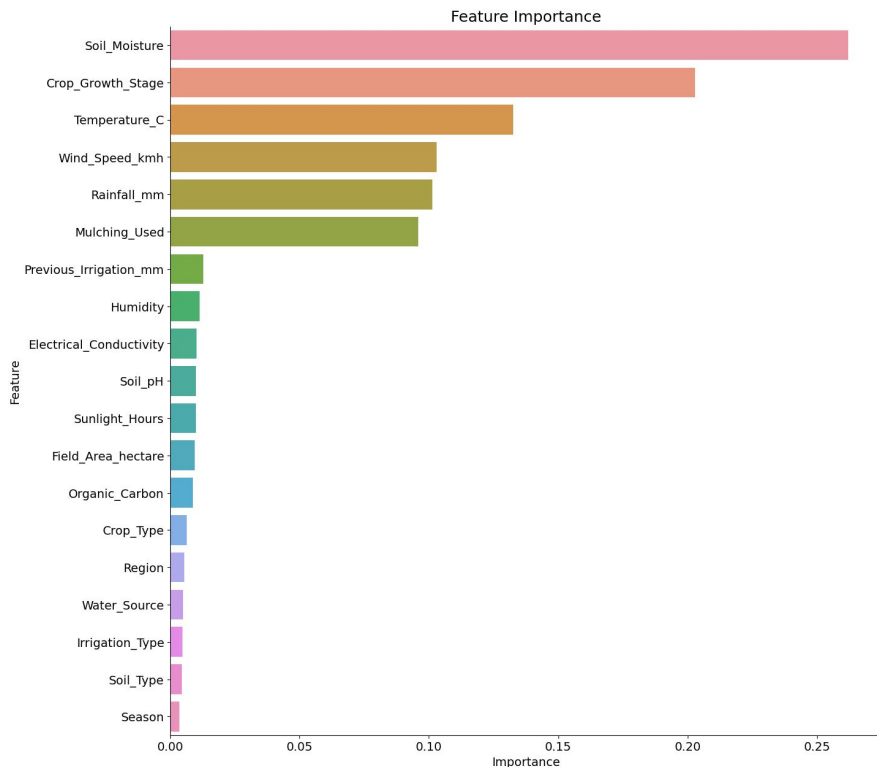
EDA: Correlation - Numerical Features



EDA: Correlation - Target Variable vs Numerical Features



EDA: Feature Importance

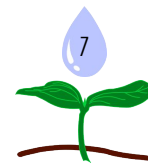


Random forest

→ with 200 estimators

→ balanced class weight

→ non-assigned value for max-depth



FEATURE ENGINEERING

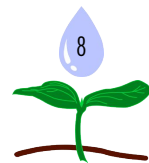
No missing values

Categorical features

→ One-hot-encoding (OneHotEncoder), drop first row

Numerical features

→ Normalization (MinMaxScaler)



EVALUATION METRIC

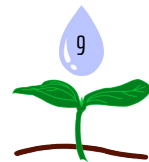
False Positives and False Negatives equally dramatic

→ **F1 score**: $2 * (\text{precision} * \text{recall} / (\text{precision} + \text{recall}))$ # punishes imbalances between precision and recall

Averaging method:

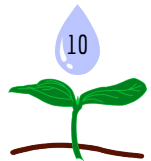
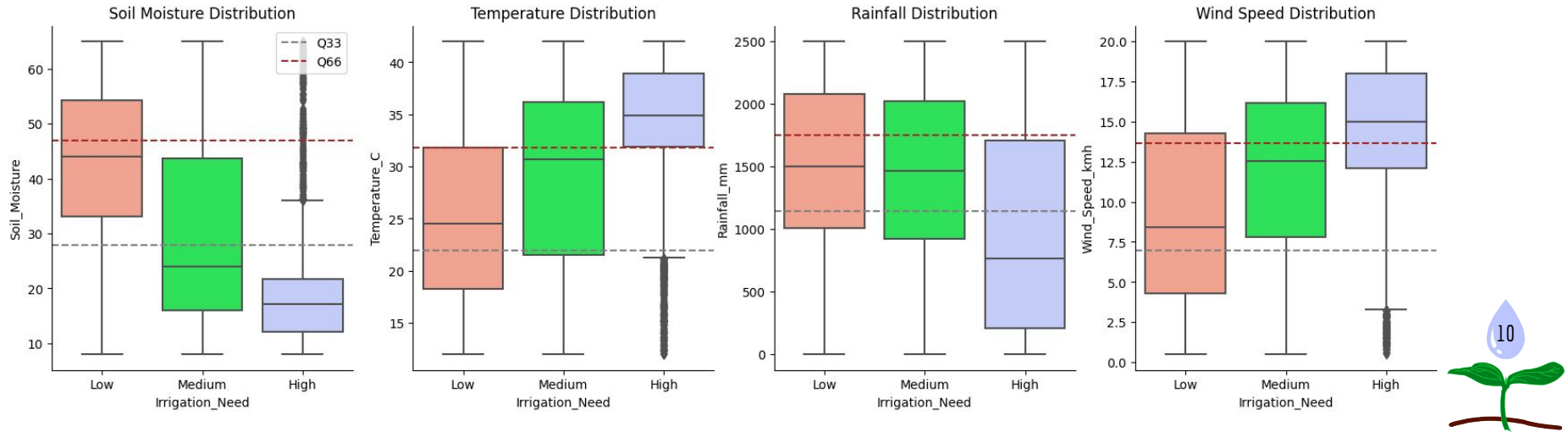
Imbalanced classes (Low - 59%, Medium - 38%, High - 3%)

→ **F1 macro**: $(F1_class0 + F1_class1 + F1_class2) / 3$ # macro → classes are treated equally
(weighted → bigger weight to larger classes)



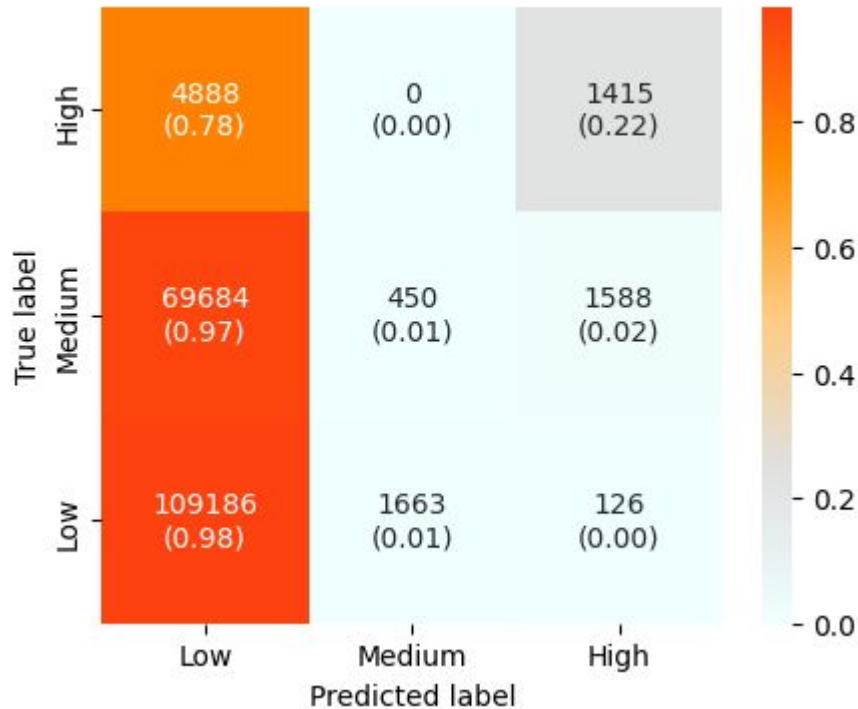
BASELINE MODEL

Simple heuristic model using domain-inspired thresholds (quantiles) to classify irrigation needs into 3 levels: Low, Medium, High.



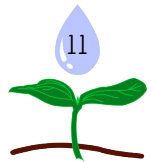
BASELINE MODEL: Evaluation

Confusion Matrix - Baseline - Validation Data



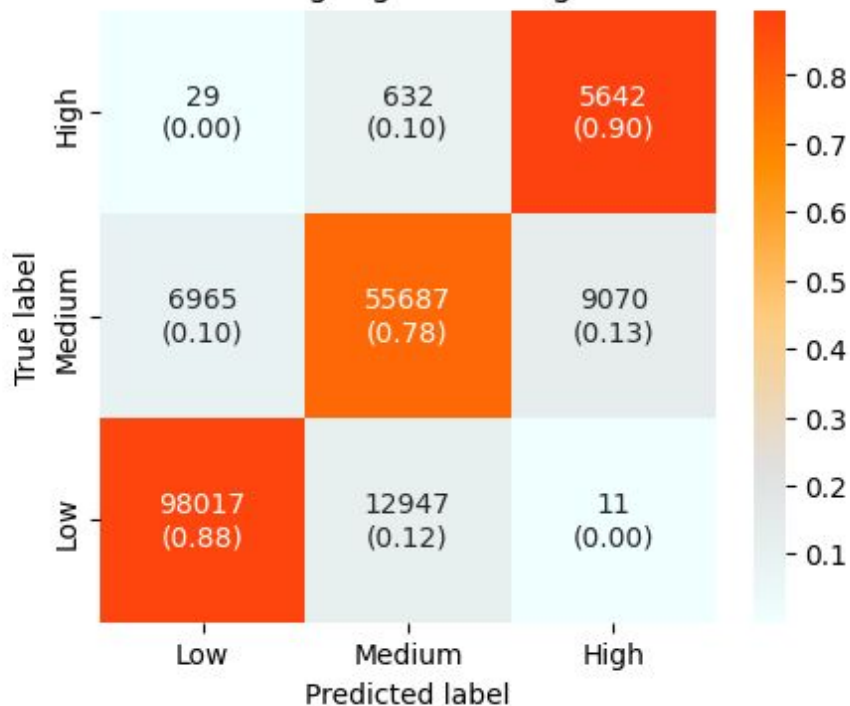
F1 macro score on training set: 0.349465

F1 macro score on validation set: 0.351049



MODEL 1: Logistic Regression

Confusion Matrix - LogReg w FeatEng - Validation Data



LogisticRegression with and without FeatEng:

'Water_Input' = 'Rainfall_mm' + 'Previous_Irrigation_mm'

'Evaporation' = 'Temperature_C' * 'Sunlight_Hours' * 'Wind_Speed_kmh'

'Moisture_Stress' = 'Temperature_C' / ('Soil_Moisture' + 1e-5)

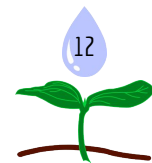
'rain_temp' = 'Rainfall_mm' * 'Temperature_C'

'humidity_temp' = 'Humidity' * 'Temperature_C'

→ Slightly better scores for model with FeatEng

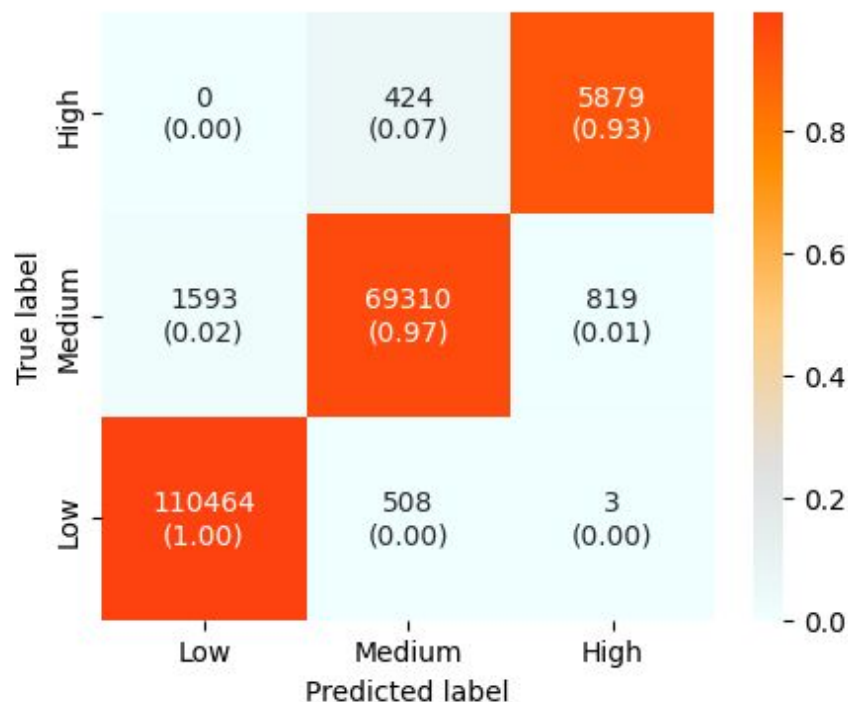
F1 macro score on training set: 0.743825

F1 macro score on validation set: 0.744749



MODEL 2: Decision Tree

Confusion Matrix - Decision Tree - Validation Data



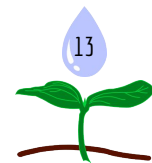
DecisionTreeClassifier with RandomizedSearch + CV

→ best parameters:

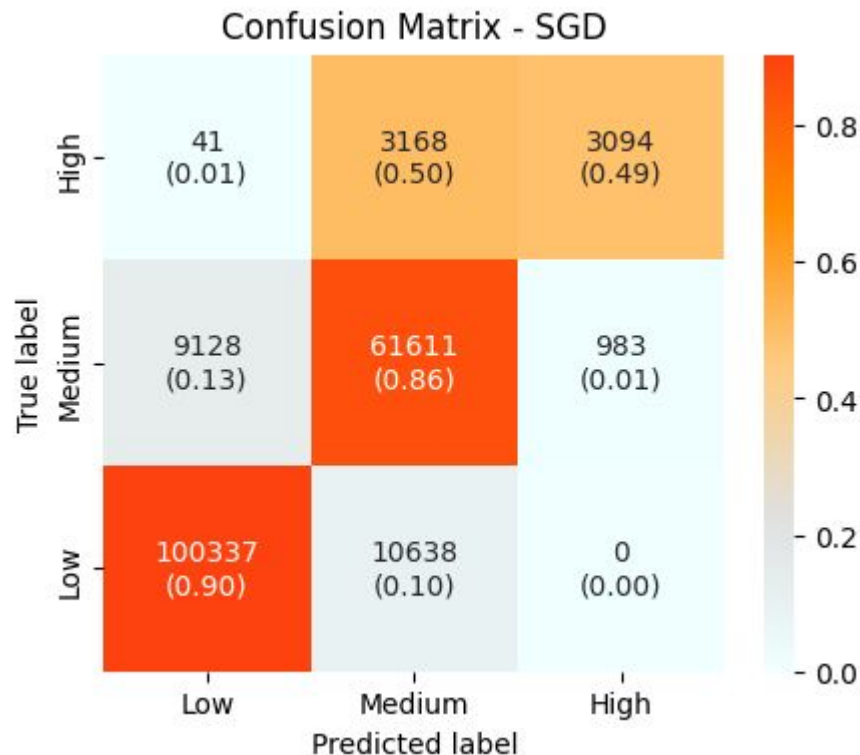
class_weight='balanced',
max_depth=7,
min_samples_split=5,
min_samples_leaf=1,
max_features=None

F1 macro score on training set: 0.958553

F1 macro score on validation set: 0.957065



MODEL 3: Stochastic Gradient Descent Classifier



SGDClassifier with RandomizedSearchCV

→ best parameters:

`class_weight='balanced',`

`eta0=0.1,`

`l1_ratio=0.5,`

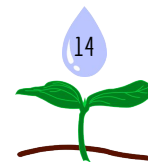
`loss='log_loss',`

`max_iter=2000,`

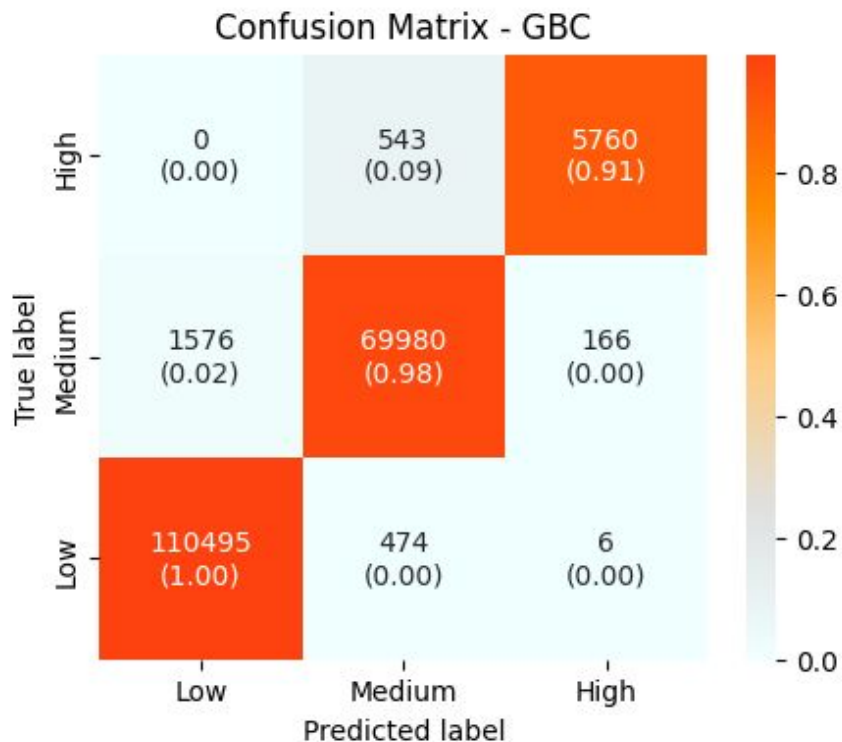
`penalty=None`

F1 macro score on training set: 0.779589

F1 macro score on validation set: 0.781255



MODEL 4: Gradient Boosting Classifier



GradientBoostingClassifier with RandomizedSearchCV

→ best parameters:

learning_rate=0.2,

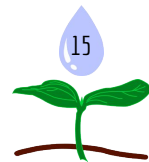
max_depth=5, max_features='sqrt',

min_samples_leaf=4, min_samples_split=10,

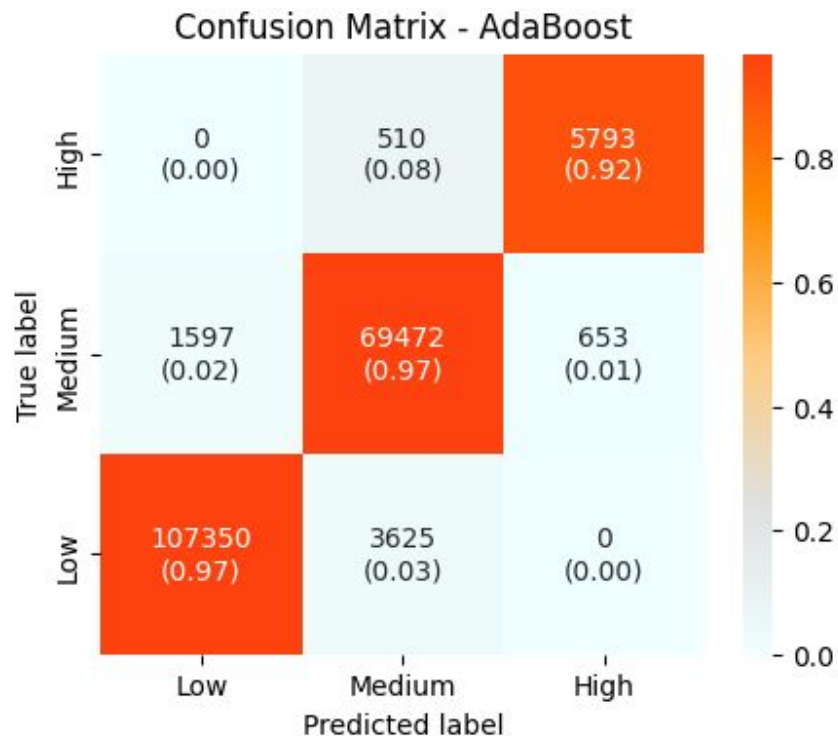
n_estimators=300

F1 macro score on training set: 0.980643

F1 macro score on validation set: 0.971004



MODEL 5: AdaBoost



AdaBoost with RandomizedSearch + CV

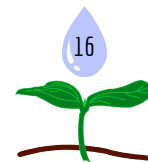
→ best parameters:

`n_estimators=120,`

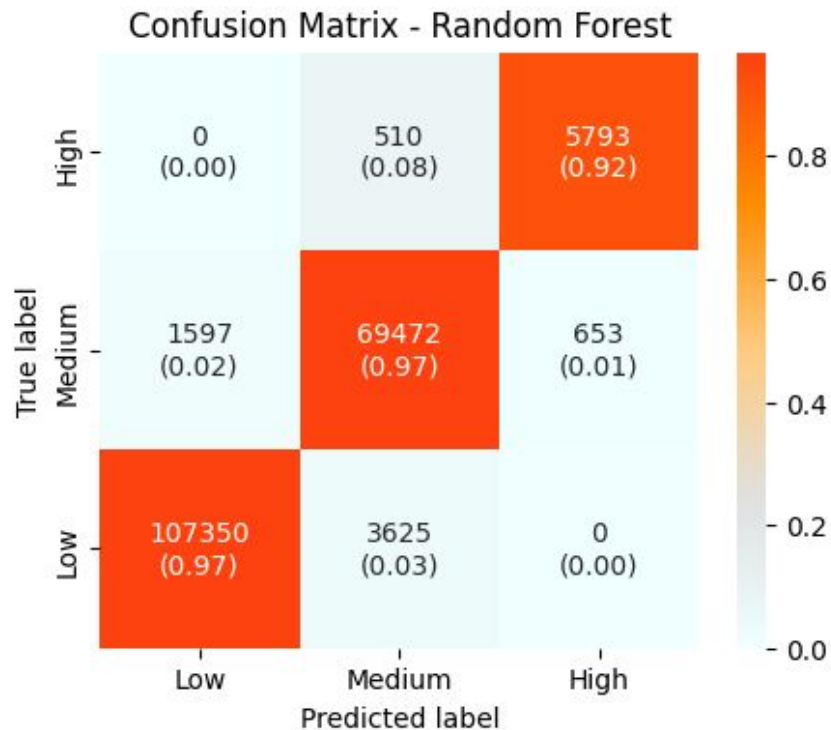
`learning_rate=1.0`

F1 macro score on training set: 0.946259

F1 macro score on validation set: 0.947032



MODEL 6: Random Forest



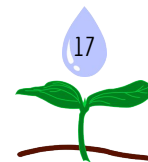
RandomForest with RandomizedSearch + CV

→ best parameters:

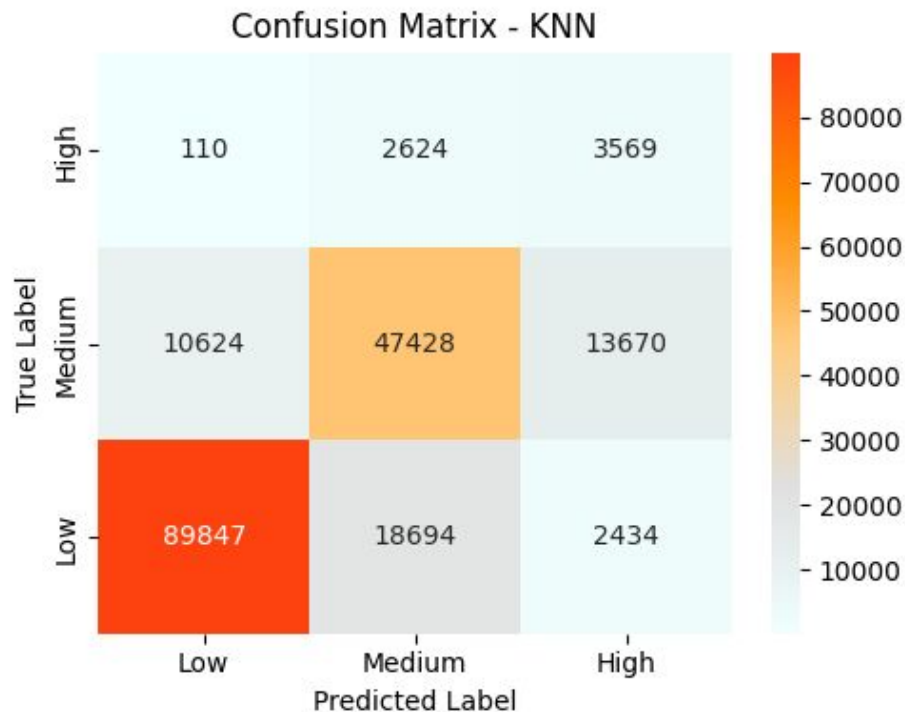
```
n_estimators=300,  
min_samples_split=10,  
min_samples_leaf=4,  
max_features=0.6,  
max_depth=35,  
bootstrap=True,  
class_weight = 'balanced',  
n_jobs=-1,  
verbose = 1
```

F1 macro score on training set: 0.985955

F1 macro score on validation set: 0.970453



MODEL 7: KNN



KNeighbourClassifier with RandomizedSearch + CV

→ best parameters:

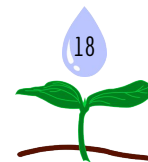
`'model__weights': 'distance',`

`'model__p': 1,`

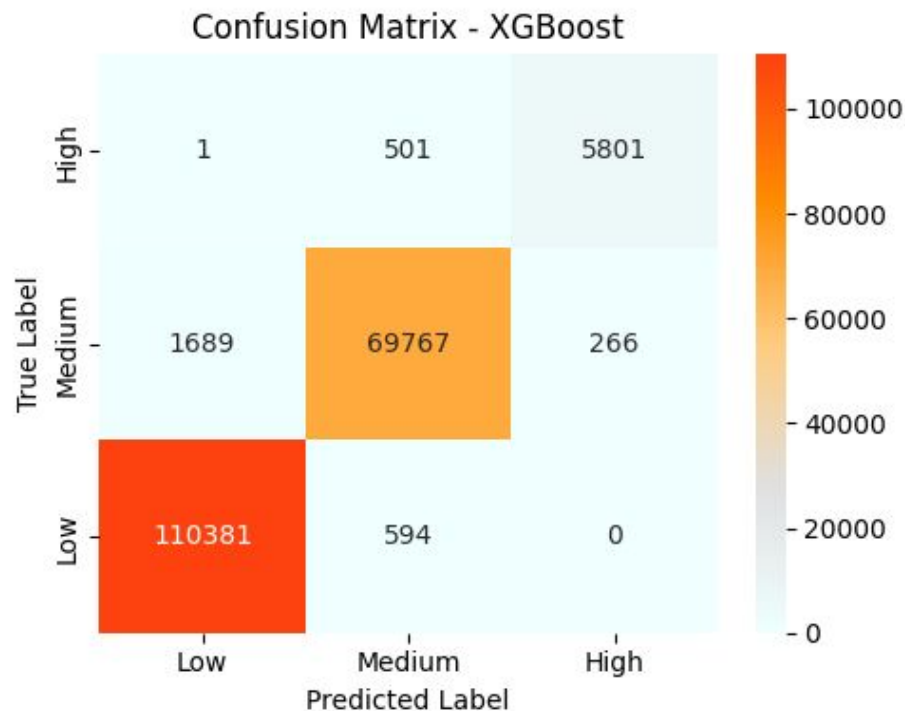
`'model__n_neighbors': 15`

F1 macro score on training set: 1.0

F1 macro score on validation set: 0.599823



MODEL 8: XGBoost



XGBoost with RandomizedSearch + CV

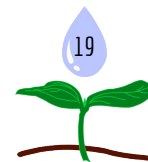
→ best parameters:

```
'model__subsample': 1.0,  
'model__n_estimators': 300,  
'model__max_depth': 5,  
'model__learning_rate': 0.1,  
'model__colsample_bytree': 0.8}
```

and SMOTE to correct for imbalanced target variable distribution

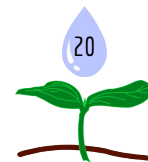
F1 macro score on training set: 0.972076

F1 macro score on validation set: 0.968761



FINAL EVALUATION: Comparison Between Models

	Logistic Regression	Decision Tree	Stochastic Gradient Descent Classifier	Gradient Boosting Classifier	AdaBoost	Random Forest	KNN	XGBoost
F1 macro (valid. set)	0.744749	0.957065	0.781255	0.971004	0.947032	0.970453	0.599823	0.968761



FINAL EVALUATION: The Chosen One

Trade-off between model simplicity and F1 score:

Comparison between Decision Tree and Gradient Boost Classifier

Best F1 score (validation set):

→ Gradient Boost Classifier: 0.971004, Decision Tree: 0.957065

Model simplicity:

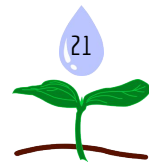
→ Decision Tree: “white-box model”, Gradient Boost Classifier: “black-box model”

Runtime:

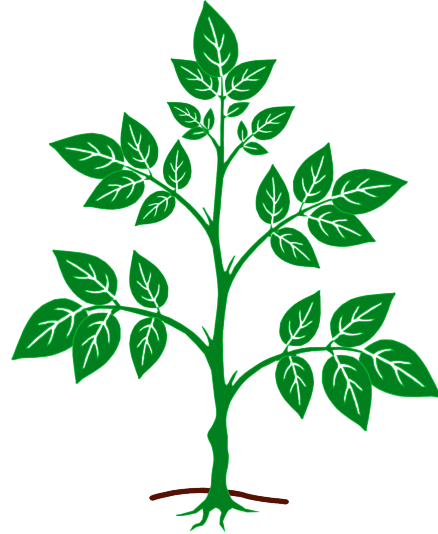
→ 1 min for Decision Tree, 154 min for Gradient Boost Classifier

Kaggle score (balanced accuracy, private score):

→ Decision Tree: 0.96415, Gradient Boost Classifier: 0.96315, (XGBoost: 0.96196, Random Forest: 0.96363)



THANK YOU



Visit our project on Github: <https://github.com/ZOZKN/ds-ml-project>

Color palette (made with Canva, based on NF orange)

Your custom palette is ready, start designing.



#FF4A11 is the neue fische orange from the preparation slides we received

